AMAN – Sprint 0 POC Results (Android)

POC 1: Values-based Blur · POC 2: Core Protection and Anti-Bypass Engine

Version 2.1 – same findings as version 2, in simpler words, with three points made clearer

Result: POC 1 – **Go with limitation** | POC 2 – **Go**

We tested on a Google Pixel 4 using real-life situations. Private tabs were also tested on a realme phone (Android 16). Testing on other brands (Samsung, Xiaomi and others) will happen in the full development phase.

"Alert only" means we can see it happen and tell the parent, but Android does not let us stop it.

NEW and UPDATED show what changed since version 1 (summary at the end). CLARIFIED shows points explained more clearly after review questions: browser image search on older Android, what the blur checks for, and what "Aman's own browser" means.

POC 2 – Core Protection and Anti-Bypass Engine

Area	What the app does	Limitation
Harmful website blocking UPDATED	Blocks harmful websites in every browser and app, on Wi-Fi and mobile data. It uses a filter that runs on the phone itself (a local VPN). The parent picks from ready-made categories: Adult, Gambling, Violence, Scams, Dating, Social and Streaming. If the child types a restricted word into a browser search, the search is closed. **Now:** the Adult category uses a real list of about **64,000 adult sites** from public sources. It also checks website names, so a new site with a clearly adult name is caught, including names in Arabic, Russian, Chinese, Korean and Thai. Violence uses a small list of gore sites. The big list is searched in a fast way, so browsing does not slow down.	Blocks whole websites, not single pages. A site opened by its number address (IP) or through a web proxy is not blocked. **Now:** about **72%** of popular adult sites are caught (before: under 1%). Small and brand-new sites are often still missed. The other categories (Gambling, Scams, Dating, Social, Streaming) still use demo lists. For launch we need a paid, licensed list that includes Arabic sites.
Safe Search	Safe Search is forced on for Google (all country sites), Bing, DuckDuckGo and YouTube (app and website). It works in every browser, including incognito. The child cannot turn it off.	Works only while our filter is running. Smaller search engines are not covered.
Block-list / Allow-list	The parent can add or remove sites, with no limit on how many. In "allow-list only" mode, only the listed sites open, but phone updates, notifications and sign-in keep working. Lists are saved on the phone. If the parent allows a site by name, it is always allowed, even if it is on the built-in adult list.	Some allowed sites may load without pictures until the site that serves those pictures is allowed too. Clearing the app's data deletes the lists (a backend would fix this).

VPN detection / restriction	Finds VPN apps and closes them when they are opened. If another VPN connects, we notice within seconds and switch it off: Android allows only one VPN at a time, so we take the slot back. The parent is alerted. Banking and government ID apps (Absher, UAE Pass and others) are left outside our filter so they keep working.	**Alert only:** another VPN set to "Always-on" in Settings, or the child removing our VPN permission.
DNS / network bypass	Changing the phone's DNS, or turning on Chrome's "Secure DNS", does not get around blocking. We check Android's Private DNS setting every 2 seconds. The parent can choose VPN filter, Private DNS or Hybrid mode (we recommend Hybrid).	**Alert only:** the child changing Private DNS in Settings. **Not blocked:** little-known private DNS servers, and other devices connected to the child's hotspot.
Internet Pause / Resume	One tap pauses or resumes the internet on the whole phone (Wi-Fi and mobile data). The child sees a clear "paused by parent" message. The pause stays on after a restart or if the app is closed. Calls and SMS still work.	Needs the VPN permission. **Alert only:** if the child removes that permission, the pause ends.
Restart and tampering	Protection starts again by itself after the phone restarts. The dashboard shows the health of protection (Active / Degraded / Not running / Tampered) and how to fix it. If permissions are turned off, we notice and alert the parent. Uninstalling the app needs an extra step.	**Alert only:** force stop, turning permissions off, uninstalling, battery saving, and the app-killing features some phone brands add. **Not possible to stop:** Safe Mode and "Clear data". NEW There is no parent PIN on the phone yet, so a child holding the phone can change Aman's own settings (see Other findings).

> **Decision needed – Device Owner mode.** If the app is set up while the phone is being reset to factory settings (by scanning a QR code), most "Alert only" items become **fully blocked**: VPN and Private DNS are locked, uninstall is blocked, and Safe Mode and hotspot are turned off. The cost is that the child's phone must be reset first. **Our recommendation:** launch as a normal Play Store app, and offer Device Owner later as an optional "maximum protection" mode.

POC 1 – Values-based Blur

Area	What the app does	Limitation

Blur inside other apps	Looks at the screen about twice a second (YouTube, Instagram, TikTok, browsers). Only the bad picture is blurred, not the whole screen. It is a real blur, the page can still be scrolled and tapped, and the covers move with the page. Everything happens on the phone: nothing is uploaded or saved.	Android asks for screen-capture permission **again every session**, so it cannot run fully on its own (the parent gets a one-tap restart message). There is a short delay (under about 1 second) before a picture is covered. Apps that block screenshots (banking, Netflix) cannot be checked. It uses a lot of battery.
Private and incognito tabs NEW	A private or incognito tab blocks screenshots. Before this change, nothing in it could be checked and nothing was covered. Now the app knows when a browser is showing a private tab, and does what the parent chose: • **Cover every picture** (default): every photo on the page is covered and the text stays readable. Covers follow the page as it scrolls. • **Don't allow private tabs:** the child is sent to the home screen with a short message. When the child goes back to a private tab, it is covered again almost at once (about 0.05 seconds). The parent's log saves only the browser name and how many pictures were covered, never the page.	Pictures in a private tab cannot be checked, so all of them are covered, safe or not. Covers are plain panels, not blur, because there is no picture we can blur. Needs the Accessibility permission and a running screen-capture session. A picture that loads while the page is scrolling can show for a moment before its cover arrives. Banking and streaming apps are still not checked, on purpose.
Parent controls UPDATED	On/off, categories (Explicit, Suggestive, Gambling, Alcohol & smoking), how strict, blur strength, and mode (bad pictures only / whole screen / cover all people). **Now also:** what to do with private tabs (cover every picture / don't allow).	The current model is good at suggestive pictures and weaker at explicit ones, so we need a model built for this job before we can promise safety. **Telling male from female is not reliable**, so we recommend "cover all people".

Browser image search CLARIFIED	In Chrome, Edge, Brave and Kiwi, every photo is covered as soon as the page opens. A photo is shown only after it has been checked and found clean. It checks for **nudity** (explicit and suggestive) and **drug images**, using models made for these. The drug check is switched on by the parent's "Explicit" setting.	**Android 14 and newer only.** Android 14 added a way to take a screenshot of just the browser window, without our own covers in it. That is what lets us check each photo cleanly. **On Android 13 and older** this feature stays off, but the phone is not left unprotected: browsers fall back to the normal "Blur inside other apps" check, the same as YouTube or Instagram. Website blocking and Safe Search work the same on every Android version, so we do not need to block install on older phones. Checks nudity and drug images only. Violence, gore, self-harm and hate symbols are **not** checked (see the list below). Photos in incognito tabs stay covered.
Aman's own browser CLARIFIED	Aman has a browser **built into the Aman app**. When the child browses inside it, each image is checked **before** it reaches the screen, so a bad image never appears. Ads are removed.	This applies only to browsing inside Aman's built-in browser, **not** to the other browsers on the phone (Chrome, Edge, Brave and others). In those other browsers, a picture is covered **after** it reaches the screen (by "Blur inside other apps" or "Browser image search" above), so it can show for a moment first.

What we changed after testing: covering the whole screen became covering only the bad picture; a dark shade became a real blur; covers that flickered now move with the page; a cover at 80% strength became full strength by using the Accessibility permission; **private tabs, which showed everything, now have every picture covered or are not allowed.**

What the blur checks for CLARIFIED

So parents do not assume full protection, this is exactly what each part checks today.

Type of content	Blur inside other apps	Browser image search	Aman's own browser
Nudity – explicit	**Yes** (on by default; current model is weaker here)	**Yes**	**Yes**
Nudity – suggestive (swimwear, underwear, revealing clothes)	**Yes** (on by default)	**Yes**	**Yes**
Drug images	**No**	**Yes** (under the "Explicit" setting)	**No**

Gambling	**Yes** (off by default; parent can turn on)	**No**	**Yes** (off by default; parent can turn on)
Alcohol & smoking	**Yes** (off by default; parent can turn on)	**No**	**Yes** (off by default; parent can turn on)
Violence and gore	**No**	**No**	**No**
Self-harm	**No**	**No**	**No**
Hate symbols	**No**	**No**	**No**

Violence-related **websites** can still be blocked by the Violence category in website blocking (POC 2). This table is only about pictures on the screen.

Permissions Needed

Permission	Used for
VPN (one-time pop-up)	Website blocking, Safe Search, lists, VPN restriction, Internet Pause
Accessibility (Settings)	Search-word blocking, full-strength blur, browser image search, **private-tab covering** UPDATED
Usage Access (Settings)	Knowing which app is open, closing VPN apps
Display over other apps	Block screen, pause screen, blur covers
Screen capture (every session)	Blur inside other apps, recognising a private tab
Notifications, Battery, Device admin	Parent alerts, staying active, making uninstall harder

Accessibility, VPN, Usage Access and seeing the full list of apps are sensitive permissions. Google Play asks for a written reason for each one, and parental control is an accepted reason.

What changed since version 1

Change	Before	Now
Adult site coverage	A demo list of a few dozen sites. Tested against the adult sites people really visit, it caught about 0.5%.	About 64,000 sites from public lists (MIT and CC BY 4.0 licences, credited in the app), plus a check on site names for new sites. Catches about **72%** of popular adult sites.

Avoiding blocking good sites	Not needed with a demo list.	The list was checked against the 10,000 most-visited websites. Entries that would block a shared server, a newspaper, a search engine and similar were removed. Phone services (updates, sign-in, notifications) can never be blocked by it. The name check skips help and recovery sites. The parent can still allow or block any site by name.
Private and incognito tabs	Could not be captured, so “Blur inside other apps” covered nothing in them while its switch still showed as on.	Recognised and handled the way the parent chose: cover every picture, or don’t allow private tabs.

Other findings

- **No parent PIN on the phone yet.** A child holding the phone can open Aman and change its settings. We need a parent PIN or lock before launch, whatever else is decided.
- **The protections already work without a backend.** Every protection runs on the phone, so the client’s “independent mode” (an older child or adult using Aman for themselves, with no parent watching) is close to what is built today. Remote control by the parent is the part still to build. See the separate *Product Modes brief* (28 Sep).
- **Still open from version 1:** a licensed site list with Arabic coverage for the other categories, a model built for image safety, and the Device Owner decision.